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Systems and methods for tracking NFT-backed instruments

Abstract

Systems, apparatuses, methods, and computer program products are disclosed for tracking NFT-backed instruments. An example method includes identifying a plurality of NFTs on a distributed blockchain ledger, associating an instrument with the cryptographic NFTs, determining a value of the cryptographic NFTs based on transactions on the distributed blockchain ledger that are associated with one or more of the cryptographic NFTs, monitoring the distributed blockchain ledger to detect transaction data of a block of the distributed blockchain ledger, the transaction data indicative of a transaction associated with a first cryptographic NFT of the cryptographic NFTs, determining a modified value of the cryptographic NFTs responsive to the transaction data, and terminating the instrument associated with the cryptographic NFTs in response to a comparison of the modified value to a predetermined threshold value indicating that the modified value is less than the predetermined threshold value.

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Background/Summary

BACKGROUND

(1) Securitized products are pools of financial instruments that produce cash flows. The financial instruments may be debt instruments like mortgages, auto loans, student loans, and credit card receivables. This process of combining the financial instruments to produce debt securities to be resold to investors is called securitization. Securitized products can provide fixed income investors with an alternative to corporate, government or municipal bonds with the potential for greater diversification and higher yields. A common example of a securitized product is a mortgage-backed security, which is secured by home or other real estate loans. Other securitized products include asset-backed securities, which are bonds created from other types of consumer debt or commercial receivables.

(2) A collateralized debt obligation (CDO) is a financial product made by pooling large sets of debt and then slicing the pool into tranches that receive interest at varying rates inversely related to the order of seniority of the tranches (an AAA tranche pays a low interest rate but is unlikely to default, while a BBB tranche pays a higher interest rate, but carries the earlier risk of loss if there are defaults in the underlying debt obligations). In doing so, investors with a variety of risk appetites may utilize a CDO to achieve their desired risk and interest rate targets.

BRIEF SUMMARY

(3) The sale prices of non-fungible tokens (NFTs) fluctuate wildly and, therefore, individual NFTs are generally not favored as an asset class to investors who are more conservative. Moreover, ownership of an NFT typically does not provide free cash flow to fund a fixed interest return. As a result, it may be difficult in conventional markets to provide investment platforms that facilitate NFT investment and/or to securitize NFTs.

(4) In contrast to these conventional techniques for securitizing financial instruments, example embodiments described herein utilize blockchain technology to securitize NFTs to provide technology support of NFT investment. To do this, embodiments according to the present disclosure may create tranches from subsets of NFTs and record and/or track each transfer of ownership of the NFTs of each subset to estimate a holistic value of that subset. The estimated value of each subset of the NFTs may be utilized to provide data structures representing financial instruments that track the subset of the NFTs. Using NFT technology rather than legacy tools for such investment tracking allows for valuation of the subset to be automatically performed. In some embodiments, the subsets of the NFTs themselves may be represented on the blockchain, allowing

for details of the subset and/or financial transactions associated with it to be transparently visible on the blockchain in immutable form.

(5) Transitioning from ad hoc legacy modes of operation to an NFT-based approach also provides an opportunity for standardization of the protocol for the management of the associated financial instruments. For example, since the interfaces to create NFTs are known and standardized, the underlying structure of the instrument associated with the NFTs can also be standardized, allowing for easier searching and management. Furthermore, an NFT-based approach enables better regulatory oversight and assists in providing clarity about the allocation of risk within the instrument associated with the NFTs. Moreover, enabling automatic valuation of pooled NFT assets in near-real-time is of heightened importance for management of NFT investments, because NFTs themselves can be transacted more quickly than the legacy assets that are traditionally the subject of securitization. Accordingly, example embodiments provide a technological platform facilitating NFT-based securitization and valuation that would not be possible using legacy securitization processes. Accordingly, the present disclosure sets forth systems, methods, and apparatuses that improve the management of financial instruments and accommodate financial instruments associated with digital structures on a blockchain.

(6) The foregoing brief summary is provided merely for purposes of summarizing some example embodiments described herein. Because the above-described embodiments are merely examples, they should not be construed to narrow the scope of this disclosure in any way. It will be appreciated that the scope of the present disclosure encompasses many potential embodiments in addition to those summarized above, some of which will be described in further detail below.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) Having described certain example embodiments in general terms above, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale. Some embodiments may include fewer or more components than those shown in the figures.

(2) FIG. **1** depicts a high-level component diagram of an illustrative example of a system architecture, in accordance with one or more aspects of the present disclosure.

(3) FIG. **2** depicts a high-level component diagram of an illustrative example of a system architecture incorporating a tranche as an NFT, in accordance with one or more aspects of the present disclosure.

(4) FIG. **3**A is a high-level component diagram of an illustrative example of a system architecture receiving a transaction related to an NFT, in accordance with one or more aspects of the present disclosure.

(5) FIG. **3**B is a high-level component diagram of an illustrative example of a system architecture modifying a value of a tranche, in accordance with one or more aspects of the present disclosure.

(6) FIG. **4** is a flow diagram of an example of a method for administering an instrument based on the value of the tranche of NFTs, in accordance with some example embodiments described herein.

(7) FIG. **5** illustrates a schematic block diagram of example circuitry embodying a device that may perform various operations in accordance with some example embodiments described herein.

(8) FIG. **6** illustrates a schematic block diagram of example circuitry embodying a device that may perform various operations in accordance with some example embodiments described herein.

(9) FIG. **7** is a flow diagram of a method for securitizing cryptographic NFTs of a blockchain, in accordance with one or more aspects of the disclosure

(10) FIG. **8** is a flow diagram of a method for generating a payment transaction based on the value of the plurality of cryptographic NFTs, in accordance with one or more aspects of the disclosure.

(11) FIG. **9** is a flow diagram of a method for creating a digital record representing a tranche in a

distributed blockchain ledger, in accordance with one or more aspects of the disclosure.

DETAILED DESCRIPTION

(12) Some example embodiments will now be described more fully hereinafter with reference to the accompanying figures, in which some, but not necessarily all, embodiments are shown. Because inventions described herein may be embodied in many different forms, the invention should not be limited solely to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements.

(13) The term “computing device” is used herein to refer to any one or all of programmable logic controllers (PLCs), programmable automation controllers (PACs), industrial computers, desktop computers, personal data assistants (PDAs), laptop computers, tablet computers, smart books, palm-top computers, personal computers, smartphones, wearable devices (such as headsets, smartwatches, or the like), and similar electronic devices equipped with at least a processor and any other physical components necessarily to perform the various operations described herein. Devices such as smartphones, laptop computers, tablet computers, and wearable devices are generally collectively referred to as mobile devices.

(14) The term “server” or “server device” is used to refer to any computing device capable of functioning as a server, such as a master exchange server, web server, mail server, document server, or any other type of server. A server may be a dedicated computing device or a server module (e.g., an application) hosted by a computing device that causes the computing device to operate as a server.

(15) The term “tranche” refers to a financial instrument backed by a segment of a pool of assets, such as NFTs. Multiple tranches may be defined for a pool of NFTs, with different ones of the tranches offering different levels of risk, maturity, or cashflow with respect to the pool. A tranche may include an attachment and a detachment point. The attachment point indicates the minimum of pool-level losses at which a given tranche begins to suffer losses. The detachment point corresponds to the amount of pool losses that completely wipe out the tranche. The riskiness of a tranche may decrease with the seniority of the tranche in the securitization of the pool. A junior tranche, for example, could have attachment and detachment points equal to 0% and 10%, respectively, of the pool exposure. Such a tranche would be intact if there are no losses but would be partly eroded with the first losses. The erosion will be complete when losses reach 10% of the pool exposure. By contrast, a mezzanine tranche with attachment and detachment points of 10% and 20%, respectively, is initially protected but would be affected as soon as losses exceed 10% of the pool size. Finally, a senior tranche with attachment and detachment points of 20% and 100% respectively will be the most protected, starting to incur losses only when both the junior and mezzanine tranches are wiped out.

(16) The term “blockchain” refers to a digital database and/or ledger that is distributed among nodes (also referred to as node devices or node computing devices) of a peer-to-peer network. In some embodiments, each of the nodes of the peer-to-peer network maintain a copy of the blockchain. The blockchain may be formed of interconnected blocks, which may represent, for example, elements of the blockchain and/or transactions associated with the elements of the blockchain. Each block contains a timestamp, transaction data, and a cryptographic hash of the previous block, forming the blockchain.

(17) The term “NFT” refers to a non-fungible token that is incorporated within blockchain. The NFT has a unique digital identifier that cannot be copied and/or substituted which may be used to certify authenticity and ownership. Transactions posted to the blockchain may record and/or change ownership of the NFT. NFTs on a blockchain may be linked to other blocks in the blockchain via a cryptographic hash, and thus may be referred to herein as a cryptographic NFT.

(18) As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(19) System Architecture

(20) Example embodiments described herein may be implemented using any of a variety of computing devices or servers. To this end, FIG. **1** depicts a high-level component diagram of an illustrative example of a system architecture **100**, in accordance with one or more aspects of the present disclosure. The system architecture **100** includes one or more administration computing devices **120** (also referred to herein as an administration device), one or more node computing devices **140** (also referred to herein as a node device).
(21) The administration device **120** and the node device(s) **140** may include one or more processing devices **160**, memory **170**, which may include volatile memory devices (e.g., random access memory (RAM)), non-volatile memory devices (e.g., flash memory) and/or other types of memory devices, storage devices **175** (which may, in some embodiments, be part of memory **170**) and one or more network interfaces **180** (also referred to herein as network hardware **180**). In certain implementations, memory **170** may be non-uniform access (NUMA), such that memory access time depends on the memory location relative to processing device **160**. It should be noted that although, for simplicity, a single processing device **160** is depicted in each of the administration device **120** and the node device(s) **140** depicted in FIG. **1**, other embodiments of the administration device **120** and the node device(s) **140** may include multiple processing devices, storage devices, or other devices. FIG. **1** and the other figures may use like reference numerals to identify like elements. A letter after a reference numeral, such as "**140**A," indicates that the text refers specifically to the element having that particular reference numeral. A reference numeral in the text without a following letter, such as "140," refers to any or all of the elements in the figures bearing that reference numeral.
(22) Processing device **160** may include a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processor implementing other instruction sets or processors implementing a combination of instruction sets. Processing device **160** may also include one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. Different ones of the administration device **120** and the node device(s) **140** may have different types of processing device **160**.
(23) Storage device **175** may comprise a distinct component from the administration device **120** and/or the node devices(s) **140**, or may comprise an element of administration device **120** and/or node devices(s) **140** (e.g., memory **170**). Storage device **175** may be embodied as one or more direct-attached storage (DAS) devices (such as hard drives, solid-state drives, optical disc drives, or the like) or may alternatively comprise one or more Network Attached Storage (NAS) devices independently connected to a communications network (e.g., network **110**). Storage device **175** may host the software executed to operate the administration device **120** and/or the node devices(s) **140**. Storage device **175** may store information relied upon during operation of the administration device **120** and/or the node devices(s) **140**, such as various computer instructions that may be used by the administration device **120** and/or the node devices(s) **140**, data and documents to be analyzed using the administration device **120** and/or the node devices(s) **140**, or the like. In addition, storage device **175** may store control signals, device characteristics, and access credentials enabling interaction between the administration device **120** and/or the node devices(s) **140**.
(24) The storage devices **175** may be embodied by any storage devices known in the art. Similarly, the administration device **120** and/or node devices(s) **140** may be embodied by any computing devices known in the art, such as desktop or laptop computers, tablet devices, smartphones, or the like. The one or more administration device **120** and/or node devices(s) **140** need not themselves be independent devices, but may be peripheral devices communicatively coupled to other computing devices.
(25) In some embodiments, the administration device **120** and/or the node device(s) **140** may be

directly or indirectly communicatively coupled through one or more of the network interfaces **180**. For example, the administration device **120** and/or the node device(s) **140** may be coupled to each other (e.g., may be operatively coupled, communicatively coupled, may communicate data/messages with each other) via network **110**. Network **110** may be a public network (e.g., the internet), a private network (e.g., a local area network (LAN) or wide area network (WAN)), or a combination thereof. In one embodiment, network **110** may include a wired or a wireless infrastructure, which may be provided by one or more wireless communications systems, such as a WIFI® hotspot connected with the network **110** and/or a wireless carrier system that can be implemented using various data processing equipment, communication towers (e.g., cell towers), etc. The network **110** may carry communications (e.g., data, message, packets, frames, etc.) between the various components of the administration device **120** and/or the node device(s) **140**.

(26) Referring to FIG. **1**, the node devices **140** may be interconnected and/or grouped into a blockchain **150**. The blockchain **150** is a linked list of records, called blocks **155**. Each of the node devices **140** may include a copy of the linked list of the blocks **155** of the blockchain **150**. For example, as illustrated in FIG. **1**, the blockchain **150** may include node devices **140**A, **140**B, **140**C, **140**D, and **140**E. The number of node devices **140** in the blockchain **150** illustrated in FIG. **1** is merely an example, and is not intended to limit the embodiments of the present disclosure. The blockchain **150** may be a public or private blockchain **150**. In a public blockchain **150**, anyone is free to join and participate in the core activities of the blockchain **150**. A private blockchain **150** allows only selected and verified participants. In some private blockchains **150**, an administrator of the blockchain **150** has the rights to override, edit, or delete entries on the blockchain **150**.

(27) The blocks **155** of the blockchain **150** are linked together, in part, using cryptography. Each block **155** contains a cryptographic hash of the previous block **155**, a timestamp, and transaction data **132** including transaction metadata. As blocks **155** each contain information about the block **155** previous to it, they form a chain, with each additional block **155** reinforcing the ones before it. Therefore, blockchains **150** are resistant to modification of their data because once recorded, the data in any given block **155** cannot be altered retroactively without altering all subsequent blocks **155**.

(28) In some embodiments, the node devices **140** may be coupled in a peer-to-peer network, where individual ones of the node devices **140** may be communicatively coupled to other ones of the node devices **140** (e.g., by network **110**). The plurality of node devices **140** may each include one or more network interfaces (e.g., similar to network interfaces **180** in FIG. **1**). In some embodiments, the blockchain **150** is provided as a distributed ledger that may be managed by the peer-to-peer network, where the node devices **140** collectively adhere to a protocol to communicate and validate new blocks **155** and record transaction data **132** therein.

(29) For example, in some embodiments, a block **155** of the blockchain **150** may include the transaction data **132** that is included when the block **155** is created (e.g., “mined”). In some embodiments, each of the transactions of the transaction data **132** may be hashed, and combinations of transactions of the transaction data **132** may be further hashed, in a merkle tree. The use of a merkle tree allows for the presence of an individual transaction in the merkle to be identified from a single merkle root hash for the block **155**. By analyzing the merkle root hash for a given block **155**, a particular transaction of the transaction data **132** may be identified that corresponds to particular elements of the blockchain **150**.

(30) The various node devices **140** of the peer-to-peer network may use consensus to validate whether a new block **155** may be added to the blockchain **150**. Consensus is used to determine whether the new block **155** is valid. For example, a consensus algorithm may be used to allow all of the node devices **140** of the blockchain **150** to reach a common agreement about the present state of the distributed ledger. In this way, consensus algorithms achieve reliability in the blockchain **150**. For example, the consensus algorithm ensures that every new block **155** that is added to the blockchain **150** is agreed upon by a subset of the node devices **140** in the blockchain **150** to be

valid. In some embodiments, the subset may be all, some, or a majority of the node devices **140** of the blockchain **150**, but is typically greater than 50%.

(31) In some embodiments, the processing device **160** of the node device **140** may execute a block validation engine **166**. For example, the block validation engine **166** may be or include computer instructions and/or circuitry configured to perform operations associated with maintaining the blockchain **150**. For example, the block validation engine **166** may include operations to perform the consensus operations described herein, and may include operations to add blocks **155** to the blockchain **150**.

(32) FIG. **1** illustrates a detailed view of one of the node devices **140** (e.g., **140**B), which is intended to explain example operations of the blockchain **150** in accordance with some embodiments of the present disclosure. The detailed view of the node device **140** and the blocks **155** of the blockchain **150** is merely schematic, and it not intended to limit the various embodiments of the disclosure.

(33) The node device **140** may include a copy of the linked list of blocks **155** of the blockchain **150**. The blockchain **150** may include a plurality of linked blocks **155**. For example, as illustrated in FIG. **1**, the blockchain **150** may include blocks **155**A, **155**B, and **155**C. The number of blocks **155** in the blockchain **150** illustrated in FIG. **1** is merely an example, and is not intended to limit the embodiments of the present disclosure.

(34) The blocks **155** of the blockchain **150** may include of the transaction data **132** as contents. The transactions of the transaction data **132** may include a plurality of operations regarding tokens of the blockchain **150**. In some embodiments, the transactions of the of the transaction data **132** may provide portions of the contents of the distributed ledger represented by the blockchain **150**. The transactions of the of the transaction data **132** may identify, for example, transfers of tokens represented by the blockchain, including creation, sale, and/or purchase of tokens and/or other operations of the blockchain **150**. In some embodiments, the transfer of tokens represented by the of the transaction data **132** may support cryptocurrency on the blockchain **150**. Cryptocurrency refers to a digital currency in which transactions are verified and records maintained by a decentralized system using cryptography, such as the blockchain **150**, rather than by a centralized authority, such as a government. For example, the transactions of the of the transaction data **132** may register the transfer of cryptocurrency between user accounts on the blockchain **150**.

(35) In some embodiments, the of the transaction data **132** of one or more of the blocks **155** of the blockchain **150** may be or include references to NFTs **138** including NFT metadata **135**. An NFT **138**, also referred to as a cryptographic NFT **138**, may include portions of the blocks **155** of the blockchain **150** including the NFT metadata **135** and an NFT asset **192**. The structure of the NFT **138** described herein, as well as the illustration of the NFT **138** in FIG. **1**, is intended to be schematic for purposes of discussion and is not intended to limit the embodiments of the present disclosure. In some embodiments, an actual physical implementation of an NFT **138** may vary from that illustrated in FIG. **1**. In some embodiments, the NFT **138** may be incorporated as a smart contract on the blockchain **150**. A smart contract may refer to a program (e.g., computer instructions) stored on the blockchain **150** that runs when predetermined conditions are met.

(36) NFT metadata **135** may include identifiers and/or other data which allows the cryptographic NFT **138** to be uniquely identifiable within the blockchain **150**. The NFT metadata **135** may be implemented according to various standards. For example, the NFT metadata **135** may be implemented according to ERC (Ethereum Request for Comment) standards ERC-721 and/or ERC-1155, though the embodiments of the present disclosure are not limited to these configurations. In FIG. **1**, each of the blocks **155** is illustrated as containing NFT metadata **135** (e.g., is part of an NFT **138**), but this is merely for purposes of illustration, and the embodiments of the present disclosure are not limited to such a configuration. In some embodiments, the blockchain **150** may include different types of blocks **155** with different varieties of transaction data **132** interspersed on the blockchain **150**.

(37) In some embodiments, the NFT metadata **135** may include an NFT link **130**. The NFT link **130** may provide a link to an NFT asset **192** being represented by the NFT metadata **135**. For example, the NFT link **130** may be a uniform resource identifier (URI) or other link referring to NFT asset **192**, which may be used to access the NFT asset **192**. In some embodiments, in addition to the NFT asset **192**, the NFT link **130** may provide a reference to additional metadata of and/or describing the NFT **138**.
(38) In some embodiments, the NFT asset **192** may be stored separately from the blockchain **150** (sometimes referred to as off-chain) though the embodiments of the present disclosure are not limited to such a configuration. In some embodiments, the NFT asset **192** may be stored in the NFT metadata **135** of the block **155**. The NET asset **192** may be any type of digital data and/or a digital reference to a physical asset. For example, the NFT asset **192** may be a digital image (e.g., digitally-created art and/or a digital representation of a photograph or art), multimedia (e.g., digital audio, video, and/or the like), and/or other digital data. For example, in some embodiments, the NFT asset **192** may be or include a portion of digital data captured from a network transaction (e.g., an email or an internet transmission) and/or other types of digital communications. In some embodiments, the NFT asset **192** may be a digital reference to a real-life asset such as real or personal property. For example, the NET asset **192** may be a reference to a physical piece of art. For purposes of example only, FIG. **1** illustrates three NFTs **138**A, **138**B, **138**C respectively having NFT metadata **135**A, **135**B, **135**C including NFT links **130**A, **130**B, **130**C to NFT assets **192**A, **192**B, **192**C.
(39) As described herein, the NFT **138** and/or the NFT metadata **135** may be uniquely identifiable within the blockchain **150**. Therefore, the NFT **138** associated with the NFT metadata **135** may be uniquely distinguished from other tokens of the blockchain **150**. The NFT **138** associated with the NFT metadata **135** may be capable of being uniquely owned. For example, the blockchain **150** may recognize a one-to-one association between the NFT **138** and a particular user account (also referred to as a “wallet”). This uniqueness may generate a market for such NFTs **138**, with demand rising or falling for the purchase and/or sale of the NFT **138** associated with the NFT metadata **135**. For example, the transactions of the of the transaction data **132** of the blockchain **150** may record transfers of ownership of the NFTs **138** associated with the NFT metadata **135** and/or NFT asset **192**.
(40) Because the NFTs **138** are hosted on the blockchain **150** (e.g., Ethereum), securitizing an NFT **138** may be difficult using legacy securitization processes. Some embodiments of the present disclosure may provide for the creation of an instrument **168** (e.g., a financial instrument **168**) that is associated with a subset **195** of cryptographic NFTs **138** on the blockchain **150**. In some embodiments, one or more of the NFTs **138** may be grouped into a subset **195** by an instrument administration engine **164** of the administration device **120**. The subset **195** may also be referred to herein as a tranche **195**. The subset **195** may combine one or more of the NFTs **138** into an asset tracked by instrument **168**. For example, FIG. **1** illustrates that a first NFT **138**A and a second NFT **138**B are grouped into a subset/tranche **195**. A third NFT **138**C, as well as its corresponding third block **155**C, are not included in the tranche **195** in the non-limiting example of FIG. **1**.
(41) The subset and/or tranche **195** may represent a segmentation of the NFTs **138** that may be provided to customers and/or investors as instrument **168**. The tranche **195** of the instrument **168** may represent a subset of the NFTs **138** having risk characteristics such as a particular term, risk level, and/or return. Different terms, risk levels, terms, and/or returns may be provided by different tranches **195**. Thus, different combinations of the NFTs **138** may be made that provide different tranches **195**, and may represent different investment products that may be associated with the instruments **168**.
(42) By representing the tranche **195** as a collection of NFTs **138** within blocks **155** on the blockchain **150**, a number of benefits may be accomplished. For example, the tranche **195** may be more easily analyzed. Each block **155** of the tranche **195** may be traversed to identify the NFT **138**

and the NFT metadata **135** analyzed to retrieve the NFT link **130** to the NFT asset **192**. Utilizing the blockchain **150** allows the NET asset **192** of each NFT **138** of the tranche **195** associated with the instrument **168** to be systematically and programmatically analyzed in a way that is not currently possible.

(43) The instrument **168** may be created and/or maintained by instrument administration engine **164**. In some embodiments, the instrument **168** may be stored on the administration device **120** in instrument store **169** of the administration device **120**, which may be, for example, a portion of the memory **170** and/or the storage device **175**, but the embodiments of the present disclosure are not limited thereto. In some embodiments, the instrument **168** may be stored separately.

(44) In some embodiments, the instrument administration engine **164** may perform operations to create, maintain, and/or terminate the instrument **168**. For example, the instrument administration engine **164** may select NFTs **138** for inclusion in the tranche **195**. In some embodiments, the instrument administration engine **164** may perform operations to provide a valuation of the instrument **168**, additional details of which will be provided herein. In some embodiments, the instrument administration engine **164** may be responsive for performing transactions (e.g., utilizing network **110**) that facilitate payments associated with the instrument **168**.

(45) The instrument **168** may allow for investments to be made in the tranche **195** of NFTs **138**. For example, as the collective value of the NFTs **138** increase, the value of the instrument **168** which represents the tranche **195** of NFTs may increase, including associated payouts for the instrument **168**. Similarly, as the collective value of the NFTs **138** decreases, the value and associated payments of the instrument **168** may decrease. In some embodiments, various risk positions may be established for the instrument **168** that allow for different payout options and/or associated risks based on the value of the tranche **195** of NFTs **138**. These risk positions may be incorporated as part of the instrument **168**. Additional details for an example of the types of instrument **168** that may be provided associated with a tranche **195** of NFTs **138** will be described herein.

(46) In some embodiments, the instrument **168** associated with the tranche **195** itself may be provided and/or recorded as an NFT **138**. For example, FIG. **2** depicts a high-level component diagram of an illustrative example of a system architecture **200** incorporating a tranche **195** as an NFT **138**, in accordance with one or more aspects of the present disclosure. A description of elements of FIG. **2** that have been previously described will be omitted for brevity.

(47) In FIG. **2**, blockchain **150** is illustrated, and blocks **155**A, **155**B, and **155**C may represent blocks **155** that represent and/or include NFTs **138** as in FIG. **1**. Blocks **155**A and **155**B have been grouped into a tranche **195** represented by instrument **168** (e.g., by instrument administration engine **164** of the administration device **120**) as described with respect to FIG. **1**. In some embodiments, the tranche **195** may also be represented by a block **155** on the blockchain **150**. For example, a fourth block **155**D may be or include reference to a fourth NFT **138**D that includes NFT metadata **135**D. The NFT metadata **135**D may identify the contents of the blockchain **150** associated with the NFT **138**D, which allows the NFT **138**D to be uniquely identifiable within the blockchain **150**. In some embodiments, the NFT metadata **135**D may include a NFT link **130**D. The NFT link **130**D may provide a link to the NFT asset, which may be instrument data **292** of the instrument **168** for the tranche **195** being represented by the NFT **138**D. For example, the NFT link **130**D may be a URI or other link referring to instrument data **292** describing the tranche **195**. In some embodiments, the NFT link **130**D may refer to a unique identification number or other unique identifying characteristic of the subset/tranche **195** and/or instrument **168** (i.e., a unique identification number that may be queried to determine the elements of the subset/tranche **195** and/or instrument **168**). In some embodiments, the instrument data **292** may be or refer to the instrument **168** stored in the instrument store **169** of the administration device **120**.

(48) The NFT link **130**D for the instrument NFT **138**D may be used to access the instrument data **292** associated with the tranche **195** and/or the instrument **168**. In some embodiments, the instrument data **292** may be stored separately from the blockchain **150** (e.g., off-chain) though the

embodiments of the present disclosure are not limited to such a configuration. In some embodiments, the instrument data **292** may be stored in the NFT metadata **135** of the block **155**. When stored off-chain, the instrument data **292** may be capable of being modified without modifying the NFT metadata **135**, which may allow for the instrument data **292** to be kept current without having to alter the block **155**.

(49) The instrument data **292** may describe the tranche **195** being represented by the respective NFT **138**D of the blockchain **150**. The instrument data **292** may include a link to the NFTs **138** that make up the tranche **195**. For example, referring to FIG. **2**, the instrument data **292** may include a link to the first NFT **138**A and the second NFT **138**B that make up the tranche **195**. In this way, the instrument data **292** may be examined to determine the respective members of the tranche **195**, which may then be traversed and analyzed (e.g., programmatically) to detect the members of the tranche **195**.

(50) The instrument data **292** may also include additional details regarding the makeup of the tranche **195**. The instrument data **292** may include data describing the tranche **195**, including risk characteristics of the tranche **195**. For example, the instrument data **292** may include an indication of the seniority level of the tranche **195** (e.g., junior, mezzanine, senior, etc.) as well as attachment and detachment points of the tranche **195**. Other data regarding the tranche **195** may be stored in the instrument data **292** without exceeding the scope of the present disclosure.

(51) Referring to FIGS. **1** and **2**, the use of an NFT **138** to represent the tranche **195** of the NFTs **138** of the instrument **168** may allow for the members of the tranche **195** to be transparently displayed on the blockchain **150**. Because the contents of the blockchain **150** are immutable and traversable, the tranche **195**, as well as the NFTs **138** that make it up, may be transparently displayed and programmatically traversed to determine the underlying blocks **155** representing the NFTs **138** of the tranche **195**, and thus the NFTs **138** representing the instrument **168** may be programmatically reviewed and/or analyzed. The blocks **155** representing the NFTs **138** of the tranche **195** may further lead to the NFT assets **192** providing additional data regarding the NFTs **138** of the instrument **168**. In this way, detailed data regarding a tranche **195** of the instrument **168**, as well as the NFTs **138** which make up the tranche **195**, may be programmatically accessed.

(52) In some embodiments, the creation of a block **155** that includes an NFT **138** may trigger its inclusion in the tranche **195**. For example, the instrument administration engine **164** may detect the creation of a block **155** having NFT metadata **135** and determine that the block **155** is acceptable for inclusion in the tranche **195**. In some embodiments, the administration device **120** may receive a request (e.g., a communication on the network **110**) for the creation of the instrument **168** and/or a tranche **195** associated with an existing instrument **168**. The request may include parameters for the tranche **195** and/or instrument **168** to be created. In response to the request, the instrument administration engine **164** may scan the blockchain **150** to detect blocks **155** that include NFT metadata **135** including an NFT link **130** (see FIG. **1**) and analyze the NFT asset **192** associated with the NFT **138** to determine that the block **155** is acceptable (e.g., the characteristics of the NET asset **192** match those of the requested tranche **195**) for inclusion in the tranche **195**.

(53) In some embodiments, the use of a blockchain **150** to represent the instruments **168** and/or the NFTs **138** of the tranche **195** may allow for the tranche **195** to be created, modified and/or administered responsive to transactions that are detected on the blockchain **150**. FIG. **3**A is a high-level component diagram of an illustrative example of a system architecture **300** detecting a transaction related to an NFT **138**, in accordance with one or more aspects of the present disclosure. A description of elements of FIG. **3**A that have been previously described will be omitted for brevity.

(54) In FIG. **3**A, blockchain **150** is illustrated having blocks **155**A, **155**B, and **155**C including respective NFTs **138**A, **138**B, and **138**C as in FIG. **1**. In some embodiments, the instrument administration engine **164** of the administration device **120** may monitor the blockchain **150** to detect the addition of particular blocks **155**. Referring to FIG. **3**A, the administration device **120**

may detect a fourth block **155**D added to the blockchain **150**. The block **155**D may include transaction data **132** that is indicative of one or more transactions associated with the blocks **155** of the blockchain **150**. Referring to FIG. **3**A, an example is illustrated in which the transaction data **132** of the block **155**D refers to the second NFT **138**B of the second block **155**B, which may be part of the tranche **195** (as illustrated in FIG. **1**). The transaction data **132** may refer, for example, to a creation and/or a sale of the second NFT **138**B that is recorded on the blockchain **150**.

(55) Responsive to detecting the creation of the block **155**D referencing the transaction data **132**, the instrument administration engine **164** of the administration device **120** may create and/or modify tranche **195**. For example, the tranche **195** may be created including the first NFT **138**A and the second NFT **138**B. As a non-limiting example, the instrument administration engine **164** may determine, based on the transaction data **132** of the block **155**D on the blockchain **150**, that the NFT **138**B referred to by the second block **155**B has a profile that is acceptable for the tranche **195**. Stated another way, the instrument administration engine **164** of the administration device **120** may determine characteristics of various NFTs **138** associated with blocks **155** on the blockchain **150** based on transaction data **132** posted to the blockchain **150**. For example, in some embodiments, the instrument administration engine **164** may be configured to calculate a risk and/or value of a particular instrument **168** (e.g., based on instrument data **292** associated with the instrument **168**, as illustrated in FIG. **2**), and may be configured to calculate an associated risk and/or value of a tranche **195** based on the NFTs **138** that make up the tranche **195**. By monitoring the blockchain **150**, the instrument administration engine **164** may be able to automatically generate the tranche **195** having the appropriate characteristics, such as risk level, term, value, etc. so as to be included in the tranche **195**. By using the blockchain **150**, decisions about the generation of a tranche **195** may be made automatically based solely on data that may be retrieved through access to the blockchain **150**.

(56) In some embodiments, the instrument administration engine **164** may be configured to calculate a value **268** of the instrument **168** based on the respective values of the NFTs **138** that make up the tranche **195** associated with the instrument **168**. For example, the value of the NFT **138** may be calculated as a sum of the respective values of the NFTs **138** that make up the tranche **195**. In some embodiments, the value of a respective NFT **138** may be based on the last recorded (e.g., recorded in the blockchain **150**) sale of the NFT **138**. For example, as NFTs **138** are bought and sold, individual values of the NFTs **138** may be adjusted. In some embodiments, the value of a particular NFT **138** may be time adjusted and/or normalized. In some embodiments, the value of a particular NFT may be normalized based on a duration between two transactions associated with the NFT **138**. For example, in some embodiments, the most two recent transactions of individual NFTs **138** can be used to estimate the current value of the NFT **138**. For example, if an NFT **138** has a transaction valued at $100 NFT in a first year and sells at $150 on June 30th of the same year, a value of the NFT **138** on December 31st of that year may be estimated to be $200. For computing the value of individual NFTs **138**, many other extrapolation or estimation methods can be used as well.

(57) FIG. **3**B is a high-level component diagram of an illustrative example of a system architecture **300** modifying a value of a tranche **195**, in accordance with one or more aspects of the present disclosure. A description of elements of FIG. **3**B that have been previously described will be omitted for brevity.

(58) In FIG. **3**B, blockchain **150** is illustrated having blocks **155**A, **155**B, and **155**C including respective NFTs **138**A, **138**B, and **138**C as in FIG. **1**. In addition, FIG. **3**B illustrates the blockchain **150** after the creation of the tranche **195** responsive to the block **155**D illustrated in FIG. **3**A. In some embodiments, the instrument administration engine **164** of the administration device **120** may monitor the blockchain **150** to detect the addition of particular blocks **155**. Referring to FIG. **3**B, the administration device **120** may detect a fifth block **155**E added to the blockchain **150**. The block **155**E may include transaction data **132** that is indicative of a transaction associated with one

or more of the NFTs **138** associated with the blocks **155** of the blockchain **150**. In FIG. **3**B, an example is illustrated in which the transaction data **132** of the block **155**E refers to the first NFT **138**A, which may be associated with the tranche **195** if the instrument **168** as illustrated in FIG. **1**. In some embodiments, the transaction data **132** of the block **155**E may refer to a transaction such as a purchase of the NFT **138**A (e.g., a change of ownership).

(59) Responsive to detecting the creation of the block **155**E referencing the transaction data **132**, the instrument administration engine **164** of the administration device **120** may modify the value **268** of the instrument **168** associated with the tranche **195** to be a modified value **268**′. For example, a value **268** associated with the tranche **195** of FIG. **3**A may be modified responsive to the transaction data **132** of the block **155**E. For example, the instrument administration engine **164** may analyze the transaction associated with the NFT **138**A to determine a value of the sale of the NFT **138**A. The value **268** of the instrument **168** may be modified to the modified value **268**′ based on the contribution of the NFT **138**A to the tranche **195**. For example, the modified value **268**′ may represent the change in the valuation of the instrument **168** based on transaction (as represented by the transaction data **132**) associated with the first NFT **138**A of the tranche **195**. By monitoring the blockchain **150**, the instrument administration engine **164** may be able to automatically modify the tranche **195** to adjust the value **268**′ of the tranche **195** based on the underlying characteristics of the transactions associated with the NFTs **138** of the tranche **195**. By using the blockchain **150**, decisions about the valuation of a tranche **195** may be made automatically based solely on data that may be retrieved through access to the blockchain **150**.

(60) In some embodiments, the instrument **168** may allow for a number of different types of investment options based on the tranche **195**. For example, different risk characteristics of the instrument **168** may be modified to adjust a risk of the instrument **168**. In some embodiments, the instrument **168** may have a term, a premium payment, an upper cutoff value, and a lower cutoff value that may be adjusted to vary a relative risk of the instrument **168** based on the value **268** of the instrument **168** calculated based on the respective values of the NFTs **138** of the tranche **195**. In some embodiments, these risk characteristics may be stored as part of the data associated with the tranche **195** and/or instrument **168**, such as part of instrument data **292** of FIG. **2**.

(61) FIG. **4** is a flow diagram of an example of a method **400** for administering the instrument **168** based on the value **268** of the tranche **195** of NFTs **138**, in accordance with some example embodiments described herein. A description of elements of FIG. **4** that have been previously described will be omitted for brevity. The operations of method **400** illustrated in FIG. **4** may, for example, be performed by administration device **120** of the system architectures **100**, **200**, **300** (described previously with reference to FIGS. **1**, **2**, **3**A, and **3**B). To perform the operations described below, the administration device **120** may utilize one or more of processing device **160**, memory **170**, network hardware **180**, instrument administration engine **164**, and/or any combination thereof. It will be understood that user interaction with the administration device **120** may occur directly via network hardware **180**, or may instead be facilitated by similar or equivalent physical componentry facilitating such user interaction.

(62) In operation **410**, a tranche **195** may be created. The tranche **195** may be a collection of one or more NFTs **138** within a blockchain **150**, as described herein with respect to FIGS. **1**, **2**, **3**A, and **3**B. In some embodiments, the tranche **195** may have a plurality of risk characteristics, including a premium payment value, a maturity date and/or term, a payout term, an upper cutoff value, and a lower cutoff value. The premium value, a maturity date, payout term, an upper cutoff value, and a lower cutoff value may establish points that control the operation of the instrument **168**. As a non-limiting example, the discussion with respect to FIG. **4** will assume an example of a premium value of 3%, a maturity date of 5 years, a payout term of a year, an upper cutoff value of $1100, and a lower cutoff value of $800.

(63) The tranche **195** may be created as part of the instrument **168**, which may be managed and/or operated on by the instrument administration engine **164** described herein. In some embodiments,

the instrument **168** may be formed between a first party and a second party. For example, the first party may be a bank and the second party may be an individual investor, but the embodiments of the present disclosure are not limited to such a configuration.

(64) In the example of FIG. **4**, for every year (e.g., the payout term) of the term of the instrument **168**, if the value **268** of the tranche **195** is above the upper cutoff value, a second party is paid the premium by a first party. This will continue until the maturity date or until the value of the tranche **195** falls below the upper cutoff value. In a given year, if the current value **268** of the tranche **195** falls below the upper cutoff value, the instrument **168** terminates, and the second party pays the first party the difference between the value **268** of the tranche **195** and the upper cutoff value, with a cap on the payout being the difference between the upper cutoff value and the lower cutoff value.

(65) In operation **420**, the blockchain **150** may be monitored for the addition of new blocks **155** related to the NFTs **138** of the tranche **195**. For example, as new blocks **155** are added, transaction data **132** of the new blocks **155** may be examined. In some embodiments, the transaction data **132** may be examined to identify a reference to an NFT **138** that is part of the tranche **195**. As described herein, the NFT **138** may have a unique identification identifier that may allow for it to be uniquely identified within the transaction data **132**. For example, a tranche **195** may be created having 2 NFTs, such as a first NFT **138**A and a second NFT **138**B similar to the example of FIG. **1**.

(66) As illustrated in operation **430**, the monitoring of the blockchain **150** as performed in operation **420** may continue until the payout term has been reached, as illustrated in operation **430**. If the payout term has not been reached (**430**: N), operation **420** may be continued. If the payout term has been reached (**430**: Y), the operations may continue with operation **440** in which a value of the tranche **195** may be calculated based on the individual values of the NFTs **138** of the tranche **195**. In an example, the monitoring of the blockchain **150** may continue for the payout term of a year (e.g., the payout term). At the end of the year, operation **440** may be performed.

(67) In operation **440**, the value of the tranche **195** may be calculated by calculating the individual values of the NFTs **138** of the tranche **195**. In some embodiments, the value of an individual NFT **138** may be based on a most recent sale of the NFT **138** (as detected by monitoring of the transaction data **132** of the blocks **155** of the blockchain **150**). In some embodiments, the value of an individual NFT **138** may be based on an average of the last X transactions involving the NFT **138**, where X may be set based on a preference of the instrument **168**. For example, the value of an individual NFT **138** may be based on an average of the last two sales of the NFT **138**. In some embodiments, the value of an individual NFT **138** may be based on a threshold number (e.g., Y) of transactions of the NFT **138** that are adjusted based on a timing of the transactions. For example, if an NFT **138** has a value at the beginning of the payout term (e.g., a year) of $100 and a transaction is detected for the NFT **138** on June 30th at $150, the value of the NFT **138** at the end of the year (e.g., December 31.sup.st) may be extrapolated and/or estimated to be $200, based on the increase in the value of $50 for the first six months of the year. The calculation of the value of the individual NFT **138** will repeat for every NFT **138** of the tranche **195**. The value **268** of the tranche **195** may be the sum of the values of the individual NFTs **138** of the tranche **195**. In will be understood that other techniques for calculating the value of individual NFTs **138** may be used without deviating from the embodiments of the present disclosure. As an example, a value of the first NFT **138**A may be calculated to be $800 and the value of the second NFT **138**B may be calculated to be $700. Therefore, the value **268** of the tranche **195** may be calculated to be $1500.

(68) In operation **450**, the value **268** of the tranche **195** may be analyzed to determine if the upper cutoff value is met. If the upper cutoff value is not met (**450**: N), the operations may continue with operation **460** in which the premium for the tranche **195** may be calculated and paid. For example, with a premium value of 3%, the premium may be calculated to be $45 based on the calculated value **268** of the tranche **195** (3%×$1500). In some embodiments, the premium may be paid by the first party to the second party.

(69) After the premium has been calculated and paid in operation **460**, the operations may continue

with operation **470** in which it is determined whether the instrument **168** has matured. If the instrument **168** has not matured (**470**: N), the operations begin again at operation **420** to monitor the blockchain **150**. If the instrument **168** has matured (**470**: Y), the instrument **168** may terminate at operation **490**. As an example, if the instrument **168** has a maturity date of five years, the monitoring of the blockchain **150** and the payout of the premium (assuming the upper cutoff value has not been bet) will continue for five years before the instrument **168** terminates.
(70) If, in operation **450**, it is determined that the value **268** of the tranche **195** is below the upper cutoff value (**450**: Y), the operations may proceed to operation **480** in which a payout is made of a minimum of the difference between the current value **268** of the tranche **195** and the difference between the upper cutoff value and the lower cutoff value. If, for example, the value of the first NFT **138**A was determined to be $500 and the value of the second NFT **138**B was determined to be $500 (for a value of the tranche **195** of $1000) then operation **450** would determine that the upper cutoff limit of $1100 was met, and an amount of the difference between the upper cutoff value and the value **268** of the tranche **195** would be paid. In the example case, the amount paid would be $100 ($1100-$1000) based on a value of the tranche **195** of $1000 and an upper cutoff value of $1100.
(71) In some embodiments, the amount paid may be capped at a difference between the upper cutoff value and the lower cutoff value. For example, if the value **268** of the tranche **195** were calculated to be $700, the most that would be paid would be $300 ($1100-$800) based on an upper cutoff value of $1100 and a lower cutoff value of $800. In some embodiments, the payout of operation **480** may be paid by the second party to the first party.
(72) After operation **480** (e.g., after the value **268** of the tranche **195** is calculated to be below the upper cutoff value), the operations may proceed to operation **490** to terminate the instrument **168** (e.g., regardless of whether the maturity date of the instrument **168** has been reached).
(73) The operations of the method **400** to administer the instrument **168** illustrated in FIG. **4** allow for an improved securitization for a digital asset, such as an NFT **138**. Since the value of a given NFT **138** may be quite volatile, the use of a tranche **195** of NFTs **138** may allow for the volatility to be diversified. Moreover, the use of operations such as those described herein with respect to FIG. **4** may provide for sophisticated risk management that allows for many configuration options with regard to changes in the value **268** of the tranche **195**. Moreover, because NFTs can be transacted more easily and quickly than traditional assets (e.g., mortgages or the like) that are subject to securitization, the operations set forth in FIG. **4** provide a technical implementation allowing for real-world management of tranches of NFTs (or other new asset classes) that may be subject to faster trading velocity. The operations of FIG. **4** are merely an example, and those of ordinary skill in the art will recognize that numerous variations are possible without deviating from the scope of the present disclosure.
(74) In some embodiments, the payments described with respect to FIG. **4** (e.g., the premium payments of operation **460** and/or the payouts of operation **480**) may be made automatically. In some embodiments, the payments described with respect to FIG. **4** may be made electronically via the blockchain **150**. For example, the blockchain **150**, as described herein with respect to FIGS. **1**, **2**, **3**A, and **3**B may support transactions (e.g., as described by transaction data **132**) involving tokens of value within the blockchain **150**. For example, the blockchain **150** may support transactions including the transfer of cryptocurrency. In some embodiments, the administration device **120** may make the payments described with respect to FIG. **4** as cryptocurrency payments on the blockchain **150**. Such an example may allow for the transactions of FIG. **4** to take place entirely within a digital environment, with all, or most, transactions supporting the instrument **168**, including payments, to be made with respect to the blockchain **150**.
(75) Example Implementing Apparatuses
(76) Administration device **120** of the system architectures **100**, **200**, **300** (described previously with reference to FIGS. **1**, **2**, **3**A, **3**B, and **4**) may be embodied by one or more computing devices

or servers, shown as apparatus **500** in FIG. **5**. As illustrated in FIG. **5**, the apparatus **500** may include processing device **160**, memory **170**, network hardware **180**, and instrument administration circuitry **510**, each of which will be described in greater detail below. While the various components are only illustrated in FIG. **5** as being connected with processing device **160**, it will be understood that the apparatus **500** may further comprises a bus (not expressly shown in FIG. **5**) for passing information amongst any combination of the various components of the apparatus **500**. The apparatus **500** may be configured to execute various operations described above in connection with FIGS. **1**, **2**, **3**A, **3**B, and **4** and below in connection with FIGS. **7-9**.

(77) The processing device **160** (and/or co-processor or any other processor assisting or otherwise associated with the processing device **160**) may be in communication with the memory **170** via a bus for passing information amongst components of the apparatus **500**. The processing device **160** may be embodied in a number of different ways and may, for example, include one or more processing devices configured to perform independently. Furthermore, the processing device **160** may include one or more processing devices configured in tandem via a bus to enable independent execution of software instructions, pipelining, and/or multithreading. The use of the term “processing device” may be understood to include a single core processor, a multi-core processor, multiple processors of the apparatus **500**, remote or “cloud” processors, or any combination thereof.

(78) The processing device **160** may be configured to execute software instructions stored in the memory **170** or otherwise accessible to the processing device (e.g., software instructions stored on a separate storage device **175**, as illustrated in FIG. **1**). In some cases, the processing device **160** may be configured to execute hard-coded functionality. As such, whether configured by hardware or software methods, or by a combination of hardware with software, the processing device **160** represents an entity (e.g., physically embodied in circuitry) capable of performing operations according to various embodiments of the present invention while configured accordingly. As another example, when the processing device **160** is embodied as an executor of software instructions, the software instructions may specifically configure the processing device **160** to perform the algorithms and/or operations described herein when the software instructions are executed.

(79) Memory **170** is non-transitory and may include, for example, one or more volatile and/or non-volatile memories. In other words, for example, the memory **170** may be an electronic storage device (e.g., a computer readable storage medium). The memory **170** may be configured to store information, data, content, applications, software instructions, or the like, for enabling the apparatus to carry out various functions in accordance with example embodiments contemplated herein.

(80) The network hardware **180** may be any means such as a device or circuitry embodied in either hardware or a combination of hardware and software that is configured to receive and/or transmit data from/to a network and/or any other device, circuitry, or module in communication with the apparatus **500**. In this regard, the network hardware **180** may include, for example, a network interface for enabling communications with a wired or wireless communication network. For example, the network hardware **180** may include one or more network interface cards, antennas, buses, switches, routers, modems, and supporting hardware and/or software, or any other device suitable for enabling communications via a network (e.g., network **110**). Furthermore, the network hardware **180** may include the processing circuitry for causing transmission of such signals to a network or for handling receipt of signals received from a network.

(81) The network hardware **180** may further be configured to provide output to a user and, in some embodiments, to receive an indication of user input. In this regard, the network hardware **180** may comprise a user interface, such as a display, and may further comprise the components that govern use of the user interface, such as a web browser, mobile application, dedicated client device, or the like. In some embodiments, the network hardware **180** may include a keyboard, a mouse, a touch

screen, touch areas, soft keys, a microphone, a speaker, and/or other input/output mechanisms. The network hardware **180** may utilize the processing device **160** to control one or more functions of one or more of these user interface elements through software instructions (e.g., application software and/or system software, such as firmware) stored on a memory (e.g., memory **170**) accessible to the processing device **160**.

(82) In addition, the apparatus **500** further comprises an instrument administration circuitry **510** that may be configured to execute the operations of the instrument administration engine **164** described herein with respect to FIGS. **1**, **2**, **3**A, **3**B, and **4**. For example, the instrument administration circuitry **510** may be configured to monitor transaction data of blocks of a blockchain, form an instrument based on one or more NFTs of the blockchain, generate and/or request generation of a block of a blockchain containing NFT metadata associated with an instrument, and/or administer the instrument based on a value of the instrument calculated from the value of the respective NFTs of the instrument, as described herein with respect to FIGS. **1**, **2**, **3**A, **3**B, and **4**. Furthermore, the instrument administration circuitry **510** may be configured to examine a provided instrument to detect and/or extract instrument data from the instrument, as described herein with respect to FIGS. **1**, **2**, **3**A, **3**B, and **4**. The instrument administration circuitry **510** may utilize processing device **160**, memory **170**, or any other hardware component included in the apparatus **500** to perform these operations, as described in connection with FIGS. **7-9** below. The instrument administration circuitry **510** may further utilize network hardware **180** to gather data from a variety of sources (e.g., the blockchain **150** and/or storage device **175**, as shown in FIG. **1**), and/or exchange data with a user, and in some embodiments may utilize processing device **160** and/or memory **170** to perform its operations.

(83) Although components of FIG. **5** are described in part using functional language, it will be understood that the particular implementations necessarily include the use of particular hardware. It should also be understood that certain of these components of FIG. **5** may include similar or common hardware. For example, the instrument administration circuitry **510** may at times leverage use of the processing device **160**, memory **170**, or network hardware **180**, such that duplicate hardware is not required to facilitate operation of this physical element of the apparatus **500** (although dedicated hardware elements may be used in some embodiments, such as those in which enhanced parallelism may be desired). Use of the terms “circuitry,” and “engine” with respect to elements of the apparatus **500** therefore shall be interpreted as necessarily including the particular hardware configured to perform the functions associated with the particular element being described. Of course, while the terms “circuitry” and “engine” should be understood broadly to include hardware, in some embodiments, the terms “circuitry” and “engine” may in addition refer to software instructions that configure the hardware components of the apparatus **500** to perform the various functions described herein.

(84) Although the instrument administration circuitry **510** may leverage processing device **160**, memory **170**, or network hardware **180** as described above, it will be understood that any of these elements of apparatus **500** may include one or more dedicated processing devices, specially configured field programmable gate array (FPGA), or application specific interface circuit (ASIC) to perform its corresponding functions, and may accordingly leverage processing device **160** executing software stored in a memory (e.g., memory **170**), or memory **170**, or network hardware **180** for enabling any functions not performed by special-purpose hardware elements. In all embodiments, however, it will be understood that the instrument administration circuitry **510** is implemented via particular machinery designed for performing the functions described herein in connection with such elements of apparatus **500**.

(85) Node device(s) **140** of the system architectures **100**, **200**, **300** (described previously with reference to FIGS. **1**, **2**, **3**A, and **3**B) may be embodied by one or more computing devices or servers, shown as apparatus **600** in FIG. **6**. As illustrated in FIG. **6**, the apparatus **600** may include processing device **160**, memory **170**, network hardware **180**, and block validation circuitry **610**,

each of which will be described in greater detail below. While the various components are only illustrated in FIG. **6** as being connected with processing device **160**, it will be understood that the apparatus **600** may further comprises a bus (not expressly shown in FIG. **6**) for passing information amongst any combination of the various components of the apparatus **600**. The apparatus **600** may be configured to execute various operations described above in connection with FIGS. **1**, **2**, **3**A, **3**B, and **4** and below in connection with FIGS. **7-9**. The processing device **160**, memory **170**, and network hardware of the apparatus **600** of FIG. **6** may be substantially similar to those discussed herein and with respect to the apparatus **500** of FIG. **5**. As such, a duplicate description thereof will be omitted.

(86) In addition, the apparatus **600** further comprises a block validation circuitry **610** that may be configured to execute the operations of the block validation engine **166** described herein with respect to FIG. **1**. For example, the block validation circuitry **610** may be configured to perform operations associated with maintaining the blockchain **150**, such as performing the consensus operations and operations to add blocks **155** to the blockchain **150**, as described herein with respect to FIG. **1**. The block validation circuitry **610** may utilize processing device **160**, memory **170**, or any other hardware component included in the apparatus **600** to perform these operations, as described in connection with FIGS. **7-9** below. The block validation circuitry **610** may further utilize network hardware **180** to gather data from a variety of sources (e.g., the blockchain **150** and/or storage device **175**, as shown in FIG. **1**), and/or exchange data with a user, and in some embodiments may utilize processing device **160** and/or memory **170** to perform its operations.

(87) Although components of FIG. **6** are described in part using functional language, it will be understood that the particular implementations necessarily include the use of particular hardware. It should also be understood that certain of these components of FIG. **6** may include similar or common hardware. For example, the block validation circuitry **610** may at times leverage use of the processing device **160**, memory **170**, or network hardware **180**, such that duplicate hardware is not required to facilitate operation of these physical elements of the apparatus **600** (although dedicated hardware elements may be used for any of these components in some embodiments, such as those in which enhanced parallelism may be desired). Use of the terms “circuitry,” and “engine” with respect to elements of the apparatus therefore shall be interpreted as necessarily including the particular hardware configured to perform the functions associated with the particular element being described. Of course, while the terms “circuitry” and “engine” should be understood broadly to include hardware, in some embodiments, the terms “circuitry” and “engine” may in addition refer to software instructions that configure the hardware components of the apparatus **600** to perform the various functions described herein.

(88) Although the block validation circuitry **610** may leverage processing device **160**, memory **170**, or network hardware **180** as described above, it will be understood that any of these elements of apparatus **600** may include one or more dedicated processing devices **160**, specially configured field programmable gate array (FPGA), or application specific interface circuit (ASIC) to perform its corresponding functions, and may accordingly leverage processing device **160** executing software stored in a memory (e.g., memory **170**), or memory **170**, or network hardware **180** for enabling any functions not performed by special-purpose hardware elements. In all embodiments, however, it will be understood that the block validation circuitry **610** is implemented via particular machinery designed for performing the functions described herein in connection with such elements of apparatus **600**.

(89) Example Operations

(90) Turning to FIGS. **7-9**, example flowcharts are illustrated that contain example operations implemented by example embodiments described herein. The operations illustrated in FIGS. **7**, **8**, and **9** may, for example, be performed by administration device **120** of the system architectures **100**, **200**, **300** (described previously with reference to FIGS. **1**, **2**, **3**A, **3**B, and **4**), which may in turn be embodied by an apparatus **500**, which is shown and described in connection with FIG. **5**. To

perform the operations described below, the apparatus **500** may utilize one or more of processing device **160**, memory **170**, network hardware **180**, instrument administration circuitry **510**, and/or any combination thereof. It will be understood that user interaction with the administration device **120** may occur directly via network hardware **180**, or may instead be facilitated by similar or equivalent physical componentry facilitating such user interaction.

(91) FIG. **7** is a flow diagram of a method **700** for securitizing cryptographic NFTs of a blockchain, in accordance with one or more aspects of the disclosure.

(92) As shown by operation **710**, the apparatus **500** includes means, such as processing device **160**, memory **170**, network hardware **180**, instrument administration circuitry **510**, or the like, for identifying a plurality of cryptographic NFTs on a distributed blockchain ledger. In some embodiments, the cryptographic NFT and the distributed blockchain ledger may be similar to the NFT **138** and blockchain **150**, respectively, discussed herein with respect to FIGS. **1**, **2**, **3**A, **3**B, and **4**.

(93) As shown by operation **720**, the apparatus **500** includes means, such as processing device **160**, memory **170**, network hardware **180**, instrument administration circuitry **510**, or the like, for associating an instrument with the plurality of cryptographic NFTs. In some embodiments, the instrument may be similar to instrument **168** discussed herein with respect to FIGS. **1**, **2**, **3**A, **3**B, and **4**. In some embodiments, the plurality of cryptographic NFTs may be similar to the tranche **195** described herein with respect to FIGS. **1**, **2**, **3**A, **3**B, and **4**.

(94) As shown by operation **730**, the apparatus **500** includes means, such as processing device **160**, memory **170**, network hardware **180**, instrument administration circuitry **510**, or the like, for determining a value of the plurality of cryptographic NFTs based on transactions on the distributed blockchain ledger that are associated with one or more of the plurality of cryptographic NFTs. In some embodiments, the transactions may be represented by transaction data **132** of a block **155** of the blockchain **150** described herein with respect to FIGS. **1**, **2**, **3**A, **3**B, and **4**. In some embodiments, the value may be represented by the value **268** described herein with respect to FIGS. **1**, **2**, **3**A, **3**B, and **4**.

(95) As shown by operation **740**, the apparatus **500** includes means, such as processing device **160**, memory **170**, network hardware **180**, instrument administration circuitry **510**, or the like, for monitoring the distributed blockchain ledger to detect transaction data of a block of the distributed blockchain ledger, the transaction data indicative of a transaction associated with a first cryptographic NFT of the plurality of cryptographic NFTs. In some embodiments, the transaction data may be similar to the transaction data **132** described herein with respect to FIGS. **1**, **2**, **3**A, **3**B, and **4**. In some embodiments, the transaction associated with the first cryptographic NFT of the plurality of cryptographic NFTs may refer to a transaction such as a purchase of the first cryptographic NFT (e.g., a change of ownership) or other transaction associated with the first cryptographic NFT on the blockchain **150**.

(96) As shown by operation **750**, the apparatus **500** includes means, such as processing device **160**, memory **170**, network hardware **180**, instrument administration circuitry **510**, or the like, for determining a modified value of the plurality of cryptographic NFTs responsive to the detection of the transaction data. In some embodiments, the modified value may be represented by the modified value **268**′ described herein with respect to FIG. **3**B. In some embodiments, determining the modified value of the plurality of cryptographic NFTs responsive to the detection of the transaction data may include determining a normalized value of the first cryptographic NFT based on a duration between the transaction data of the block and prior transaction data associated with the first cryptographic NFT. In some embodiments, determining the modified value of the plurality of cryptographic NFTs may be performed at periodic intervals.

(97) As shown by operation **760**, the apparatus **500** includes means, such as processing device **160**, memory **170**, network hardware **180**, instrument administration circuitry **510**, or the like, for terminating the instrument associated with the plurality of cryptographic NFTs in response to a

comparison of the modified value to a predetermined threshold value indicating that the modified value is less than the predetermined threshold value. In some embodiments, the predetermined threshold value may be represented by the upper cutoff value described herein with respect to FIG. **4**.
(98) In some embodiments, the method **700** may further include creating a digital record on the distributed blockchain ledger representing the instrument associated with the plurality of cryptographic NFTs. In some embodiments, the digital record may be similar to the NFT **138** created in association with an instrument **168**, as described herein with respect to FIG. **2**. In some embodiments, the digital record comprises metadata associated with the instrument, the metadata comprising a plurality of risk characteristics of the instrument associated with the plurality of cryptographic NFTs. In some embodiments, the plurality of risk characteristics include a term of the instrument, a premium payment associated with the instrument, an upper cutoff value for the instrument, and a lower cutoff value for the instrument.
(99) In some embodiments, the method **700** may further include determining a payment based on the modified value of the plurality of cryptographic NFTs and generating a transaction on the distributed blockchain ledger to represent the payment. In some embodiments, the payment may represent a transfer of cryptocurrency on the distributed blockchain ledger.
(100) FIG. **8** is a flow diagram of a method **800** for generating a payment transaction based on the value of the plurality of cryptographic NFTs, in accordance with one or more aspects of the disclosure.
(101) As shown by operation **810**, the apparatus **500** includes means, such as processing device **160**, memory **170**, network hardware **180**, instrument administration circuitry **510**, or the like, for generating a current value of a plurality of cryptographic NFTs that are associated with an instrument, wherein the instrument comprises an upper cutoff value, a lower cutoff value, and a premium value. In some embodiments, the plurality of cryptographic NFTs may be similar to the tranche **195** of NFTs **138** of a blockchain **150**, as described herein with respect to FIGS. **1** to **4**. The instrument may be similar to instrument **168**, which may be a financial instrument, as described herein with respect to FIGS. **1** to **4**.
(102) As shown by operation **820**, the apparatus **500** includes means, such as processing device **160**, memory **170**, network hardware **180**, instrument administration circuitry **510**, or the like, for comparing the current value of the plurality of cryptographic NFTs to the upper cutoff value. In some embodiments, the current value of the plurality of cryptographic NFTs may be a sum of the current value of respective ones of the plurality of cryptographic NFTs. In some embodiments, the current value of a cryptographic NFT **138** may be based on transactions on the blockchain **150** associated with the cryptographic NFT **138**. In some embodiments, the value of the cryptographic NFT **138** may be based on a defined number of sales of the cryptographic NFT **138** that are recorded on the blockchain **150**, which may be determined based on transaction data **132** of a block **155** of the blockchain **150**.
(103) As shown by operation **830**, the apparatus **500** includes means, such as processing device **160**, memory **170**, network hardware **180**, instrument administration circuitry **510**, or the like, for generating, responsive to the value of the plurality of cryptographic NFTs being less than the upper cutoff value, a payment transaction for the lesser of a difference between the current value and the upper cutoff value and a difference between the upper cutoff value and the lower cutoff value. In some embodiments, the payment transaction may be a transaction on the blockchain **150**, such as a cryptocurrency transaction.
(104) As shown by operation **840**, the apparatus **500** includes means, such as processing device **160**, memory **170**, network hardware **180**, instrument administration circuitry **510**, or the like, for generating, responsive to the value of the plurality of cryptographic NFTs being equal to or greater than the upper cutoff value, a payment transaction for the premium value. In some embodiments, the payment transaction may be a transaction on the blockchain **150**, such as a cryptocurrency

transaction.

(105) FIG. **9** is a flow diagram of a method **900** for creating a digital record representing a tranche in a distributed blockchain ledger, in accordance with one or more aspects of the disclosure.

(106) As shown by operation **910**, the apparatus **500** includes means, such as processing device **160**, memory **170**, network hardware **180**, instrument administration circuitry **510**, or the like, for identifying a plurality of cryptographic NFTs on a distributed blockchain ledger. In some embodiments, the plurality of cryptographic NFTs may be similar to the tranche **195** of NFTs **138** of a blockchain **150**, as described herein with respect to FIGS. **1** to **4**.

(107) As shown by operation **920**, the apparatus **500** includes means, such as processing device **160**, memory **170**, network hardware **180**, instrument administration circuitry **510**, or the like, for creating a grouping of the plurality of cryptographic NFTs, the grouping corresponding to a tranche representing an instrument associated with the plurality of cryptographic NFTs. In some embodiments, the tranche may be similar to tranche **195** discussed herein with respect to FIGS. **1-4**. The instrument may be similar to instrument **168**, which may be a financial instrument, as described herein with respect to FIGS. **1** to **4**. In some embodiments, creating the grouping may include compiling information related to the individual cryptographic NFTs of the plurality of cryptographic NFTs, including identifying information, such as the unique identification numbers of the blocks of the blockchain corresponding to the individual cryptographic NFTs of the plurality of cryptographic NFTs.

(108) As shown by operation **930**, the apparatus **500** includes means, such as processing device **160**, memory **170**, network hardware **180**, instrument administration circuitry **510**, or the like, for creating a digital record on the distributed blockchain ledger representing the instrument. In some embodiments, the digital record may include metadata associated with the tranche representing the plurality of cryptographic NFTs. The metadata associated with the tranche may be similar to NFT metadata **135** of block **155**D containing a NFT link **130**, as discussed herein with respect to FIG. **2**. In some embodiments, the NFT link **130** may refer to instrument data, such as instrument data **292** illustrated in FIG. **2**. In some embodiments, the instrument data **292** may include risk characteristics associated with the instrument. For example, in some embodiments, the instrument data may include one or more of a term of the instrument, a premium payment associated with the instrument, an upper cutoff value for the instrument, or a lower cutoff value for the instrument.

(109) FIGS. **7**, **8**, and **9** illustrate operations performed by apparatuses, methods, and computer program products according to various example embodiments. It will be understood that each flowchart block, and each combination of flowchart blocks, may be implemented by various means, embodied as hardware, firmware, circuitry, and/or other devices associated with execution of software including one or more software instructions. For example, one or more of the operations described above may be embodied by software instructions. In this regard, the software instructions which embody the procedures described above may be stored by a memory of an apparatus employing an embodiment of the present invention and executed by a processor of that apparatus. As will be appreciated, any such software instructions may be loaded onto a computing device or other programmable apparatus (e.g., hardware) to produce a machine, such that the resulting computing device or other programmable apparatus implements the functions specified in the flowchart blocks. These software instructions may also be stored in a computer-readable memory that may direct a computing device or other programmable apparatus to function in a particular manner, such that the software instructions stored in the computer-readable memory produce an article of manufacture, the execution of which implements the functions specified in the flowchart blocks. The software instructions may also be loaded onto a computing device or other programmable apparatus to cause a series of operations to be performed on the computing device or other programmable apparatus to produce a computer-implemented process such that the software instructions executed on the computing device or other programmable apparatus provide operations for implementing the functions specified in the flowchart blocks.

(110) The flowchart blocks support combinations of means for performing the specified functions and combinations of operations for performing the specified functions. It will be understood that individual flowchart blocks, and/or combinations of flowchart blocks, can be implemented by special purpose hardware-based computing devices which perform the specified functions, or combinations of special purpose hardware and software instructions.
(111) In some embodiments, some of the operations described above in connection with FIGS. **7-9** may be modified or further amplified. Furthermore, in some embodiments, additional optional operations may be included. Modifications, amplifications, or additions to the operations above may be performed in any order and in any combination.
(112) Embodiments of the present disclosure may provide improved methods and apparatus for the securitization of NFTs, including the generations of tranches therefrom. By generating instruments based on NFTs, the instruments may be represented in a way that is transparent (e.g., may be audited by viewing the blockchain) and immutable (e.g., may not be changed once added to the blockchain). The use of the blockchain allows the instruments to be programmatically searched by accessing the blockchain in a way that is not currently possible. Moreover, embodiments of the present disclosure also provide for the representation of tranches on the blockchain using NFTs. Such representation of tranches allows for the tranche to be transparently accessible and allows for the individual products that make up the tranche to be traversed. This allows for the programmatic discovery of the elements of the tranche.
(113) Embodiments of the present disclosure provide an improvement to the technology associated with blockchain processing, including the ability to securitize NFTs within a tranche. For example, the embodiments of the present disclosure reduce an amount of resources needed to track the inventory of a tranche and provide technical solutions that are not currently possible with respect to identifying elements of a tranche, including their details. Transitioning to instruments that are based on NFTs, including the use of an NFT to represent the instrument, also provides an opportunity for standardization of the protocol for the management of financial instruments. For example, since the interfaces to create NFTs are known and standardized, the underlying structure of the securitized products and/or tranches can also be standardized, allowing for easier searching and management. Furthermore, an NFT-based approach enables better regulatory oversight and assists in providing clarity into how risk and instruments are allocated within a given tranche. Accordingly, the present disclosure sets forth systems, methods, and apparatuses that accommodates the tranching of NFTs into various financial instruments and improves the management of financial instruments.
CONCLUSION
(114) Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe example embodiments in the context of certain example combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A method comprising: identifying a plurality of cryptographic non-fungible tokens (NFTs) on a distributed blockchain ledger; associating, by instrument administration circuitry, an instrument with the plurality of cryptographic NFTs; determining, by the instrument administration circuitry, a value of the plurality of cryptographic NFTs based on one or more sets of transaction data on the distributed blockchain ledger that are associated with one or more of the plurality of cryptographic NFTs; monitoring, by the instrument administration circuitry, the distributed blockchain ledger to detect transaction data of a block of the distributed blockchain ledger, the transaction data indicative of a transaction associated with a first cryptographic NFT of the plurality of cryptographic NFTs; automatically determining, by the instrument administration circuitry, a modified value of the plurality of cryptographic NFTs responsive to detection of the transaction data; automatically terminating the instrument associated with the plurality of cryptographic NFTs in response to a comparison of the modified value to a predetermined threshold value indicating that the modified value is less than the predetermined threshold value; and creating a digital record on the distributed blockchain ledger representing the instrument associated with the plurality of cryptographic NFTs, wherein the digital record comprises metadata associated with the instrument, the metadata comprising a plurality of risk characteristics of the instrument associated with the plurality of cryptographic NFTs.
2. The method of claim 1, wherein the plurality of risk characteristics comprises one or more of a term of the instrument, a premium payment associated with the instrument, an upper cutoff value for the instrument, or a lower cutoff value for the instrument.
3. The method of claim 1, wherein automatically determining the modified value of the plurality of cryptographic NFTs responsive to the detection of the transaction data comprises determining a normalized value of the first cryptographic NFT based on a duration between the transaction data of the block and prior transaction data associated with the first cryptographic NFT.
4. The method of claim 1, wherein automatically determining the modified value of the plurality of cryptographic NFTs is performed at periodic intervals.
5. The method of claim 1, further comprising: determining a payment based on the modified value of the plurality of cryptographic NFTs; and generating a transaction within a block of the distributed blockchain ledger to represent the payment.
6. The method of claim 1, wherein the plurality of risk characteristics comprises an upper cutoff value for the instrument.
7. An apparatus comprising an instrument administration circuitry, the instrument administration circuitry comprising: a memory; and a processing device, operatively coupled to the memory, to: identify a plurality of cryptographic non-fungible tokens (NFTs) on a distributed blockchain ledger; associate an instrument with the plurality of cryptographic NFTs; determine a value of the plurality of cryptographic NFTs based on one or more sets of transaction data on the distributed blockchain ledger that are associated with one or more of the plurality of cryptographic NFTs; monitor the distributed blockchain ledger to detect transaction data of a block of the distributed blockchain ledger, the transaction data indicative of a transaction associated with a first cryptographic NFT of the plurality of cryptographic NFTs; automatically determine a modified value of the plurality of cryptographic NFTs responsive to the detection of the transaction data; automatically terminate the instrument associated with the plurality of cryptographic NFTs in response to a comparison of the modified value to a predetermined threshold value indicating that the modified value is less than the predetermined threshold value; and create a digital record on the distributed blockchain ledger representing the instrument associated with the plurality of cryptographic NFTs, wherein the digital record comprises metadata associated with the instrument, the metadata comprising a plurality of risk characteristics of the instrument associated with the plurality of cryptographic NFTs.
8. The apparatus of claim 7, wherein the plurality of risk characteristics comprises one or more of a term of the instrument, a premium payment associated with the instrument, an upper cutoff value

for the instrument, or a lower cutoff value for the instrument.

9. The apparatus of claim 7, wherein, to automatically determine the modified value of the plurality of cryptographic NFTs responsive to the detection of the transaction data, the processing device is to determine a normalized value of the first cryptographic NFT based on a duration between the transaction data of the block and prior transaction data associated with the first cryptographic NFT.

10. The apparatus of claim 7, wherein the processing device is to automatically determine the modified value of the plurality of cryptographic NFTs at periodic intervals.

11. The apparatus of claim 7, wherein the processing device is further to: determine a payment based on the modified value of the plurality of cryptographic NFTs; and generate a transaction within a block of the distributed blockchain ledger to represent the payment.

12. A non-transitory computer-readable storage medium comprising instructions that, when executed by a processing device, cause the processing device to: identify a plurality of cryptographic non-fungible tokens (NFTs) on a distributed blockchain ledger; associate an instrument with the plurality of cryptographic NFTs; determine a value of the plurality of cryptographic NFTs based on one or more sets of transaction data on the distributed blockchain ledger that are associated with one or more of the plurality of cryptographic NFTs; monitor the distributed blockchain ledger to detect transaction data of a block of the distributed blockchain ledger, the transaction data indicative of a transaction associated with a first cryptographic NFT of the plurality of cryptographic NFTs; automatically determine a modified value of the plurality of cryptographic NFTs responsive to the detection of the transaction data; automatically terminate the instrument associated with the plurality of cryptographic NFTs in response to a comparison of the modified value to a predetermined threshold value indicating that the modified value is less than the predetermined threshold value; and create a digital record on the distributed blockchain ledger representing the instrument associated with the plurality of cryptographic NFTs, wherein the digital record comprises metadata associated with the instrument, the metadata comprising a plurality of risk characteristics of the instrument associated with the plurality of cryptographic NFTs.

13. The non-transitory computer-readable storage medium of claim 12, wherein the plurality of risk characteristics comprises one or more of a term of the instrument, a premium payment associated with the instrument, an upper cutoff value for the instrument, or a lower cutoff value for the instrument.

14. The non-transitory computer-readable storage medium of claim 12, wherein, to automatically determine the modified value of the plurality of cryptographic NFTs responsive to the detection of the transaction data, the processing device is to determine a normalized value of the first cryptographic NFT based on a duration between the transaction data of the block and prior transaction data associated with the first cryptographic NFT.

15. The non-transitory computer-readable storage medium of claim 12, wherein the instructions, when executed by the processing device, further cause the processing device to: determine a payment based on the modified value of the plurality of cryptographic NFTs; and generate a transaction within a block of the distributed blockchain ledger to represent the payment.